

11: Retrieving the record from database

Retrieving data from a database using JDBC is a read-only operation that fetches records without altering any rows in the table. This is commonly done using the SELECT SQL statement.

- `executeQuery()` method is used to execute SELECT statements and retrieve data from the database.
- The `executeQuery()` method returns a `ResultSet` object, which holds all the records fetched from the database.

👉 Steps For Retrieving the Data:

1. Create an SQL SELECT Statement:

```
String sql = "SELECT * FROM studentInfo";
```

2. Execute the SQL Statement:

Use the Statement object's `executeQuery()` method to execute the SELECT statement.

```
ResultSet rs = statement.executeQuery(sql);
```

3. Process the ResultSet:

- The `ResultSet` object contains all the records fetched from the database. You can iterate through the `ResultSet` using a loop to access each record.
- Data from each record can be fetched using methods that start with `get` followed by the data type of the column (e.g., `getInt()`, `getString()`).
- Either the column index or name as a parameter in the method.

```
while(rs.next()){  
    //System.out.println(rs.getInt(1) + "" + rs.getString(2) + "" + rs.getInt(3)  
    + "" + rs.getString(4));  
    System.out.println(rs.getInt("id") + "" + rs.getString("sname") + "" +  
    rs.getInt("sage") + "" + rs.getString("scity"));  
}
```

4. Close the ResultSet:

After processing the data, it's essential to close the `ResultSet` and other resources.

```
rs.close();
```

Code:

```
import java.sql.*;

public class RetrieveJdbc {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/yourDatabaseName";

        String username = "yourUsername";

        String password = "yourPassword";

        String sql = "SELECT * FROM studentInfo";

        Connection conn = null;

        Statement stmt = null;

        ResultSet rs = null;

        try {

            conn = DriverManager.getConnection(url, username, password);

            stmt = conn.createStatement();

            rs = stmt.executeQuery(sql);

            while (rs.next()) {

                System.out.println(rs.getInt("id") + " " + rs.getString("sname") + " " +
rs.getInt("sage") + " " + rs.getString("scity"));

            }

        }

    }

}
```

```
    } catch (SQLException e) {  
        e.printStackTrace();  
    } finally {  
        try {  
            if (rs != null) rs.close();  
            if (stmt != null) stmt.close();  
            if (conn != null) conn.close();  
        } catch (SQLException e) {  
            e.printStackTrace();  
        }  
    }  
}  
}
```